What Becomes of Geometry When the Manifold Is Gone?

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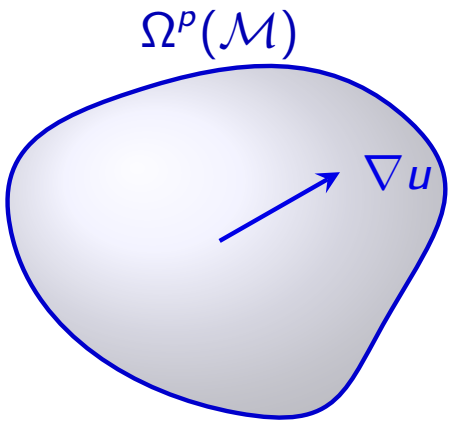
Goal: describe geometric notions intrinsically on a finite combinatorial structure, not as a discretisation step.

Roadmap

- **Combinatorial differential forms on a cell complex:** representation of physical quantities.
- **Cochains and operations on a quasi-cubical complex:** the algebraic core.
- **Scalar transport:** conservation and weak formulations as cochain identities + adjoints.
- **Vector-valued forms:** varying fibres, transport along edges, and holonomy.
- **Obstructions:** curvature (holonomy defect), and torsion-like defects (compatibility).

The question

- Differential geometry usually begins with a smooth manifold and fixed background structure.
- Many systems evolve by *changing adjacency/incidence* (splitting, merging, rewiring): the background is not stable.
- **Question:** what are the intrinsic analogues of forms, connections, curvature when the primary object is combinatorial?



Manifold $\mathcal{M}$

The combinatorial structure $\mathcal{M}$

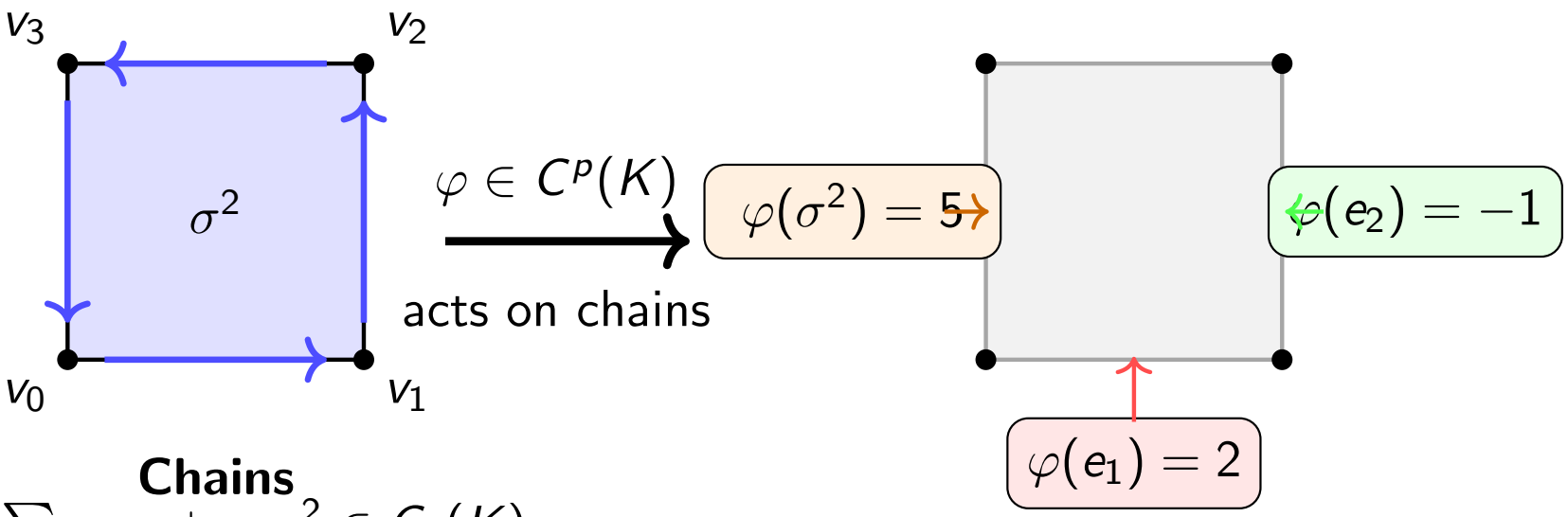
- We begin with a **finite cell complex** $\mathcal{M}$ as the primary "space" describing the relational structure of a physical system (no embedding required).
- Cells represent entities of different dimension:
  - vertices (0–cells)
  - edges (1–cells)
  - faces (2–cells)
  - higher–dimensional cells if needed
- Physical quantities are represented as **combinatorial differential forms** (Forman): a k–form assigns values to *incident pairs* of cells $\sigma^p \prec \tau^{p+k}$.
- These objects admit an **exterior derivative** defined through cell incidence.

The computational complex $\mathcal{K}$ ▸ Forman 2002

- We assume $\mathcal{M}$ is built from **simple polytopes:** each vertex is incident to exactly D $(D-1)$–cells.
- We pass to its **Forman subdivision** K:
  - vertices of $\mathcal{K}$ correspond to cells of $\mathcal{M}$,
  - adjacency reflects cell incidence in $\mathcal{M}$.
- The resulting complex $\mathcal{K}$ is **quasi-cubical**: local neighborhoods resemble products of intervals.
- Under this construction:
  - **combinatorial differential forms on $\mathcal{M}$ correspond naturally to cochains on $\mathcal{K}$.**
  - **exterior derivatives of forms on $\mathcal{M}$ correspond naturally to coboundary operators of cochains on $\mathcal{K}$.**
  - the quasi-cubical structure enables a **natural cup product on cochains**.

Chains and cochains

- $C_p(\mathcal{K})$: formal linear combinations of oriented p-cells.
- $C^p(\mathcal{K})$: linear functionals on $C_p(\mathcal{K})$.



Chains
$c = \sum_i a_i e_i + a_5 \sigma^2 \in C_p(K)$

Cochains
algebraic representation of forms

Cup product on cochains

- The cochain spaces $C^\bullet(\mathcal{K})$ carry a graded product

$$\smile : \; C^p(\mathcal{K}) \times C^q(\mathcal{K}) \longrightarrow C^{p+q}(\mathcal{K}),$$

  called the **cup product**. ▸ Berbatov 2022

- Under the correspondence between combinatorial differential forms on $\mathcal{M}$ and cochains on $\mathcal{K}$, the cup product is the algebraic counterpart of the **exterior (wedge) product** of forms.

- This is needed later for **higher-order algebraic constructions**, including constitutive couplings, vector-valued operations, and bracket-type structures.

- NOTE: Unlike the smooth wedge product, the cup product is in general **not associative**.

Boundary and coboundary operators

- Boundary operator on chains:

$$\partial : C_{p+1}(\mathcal{K}) \to C_p(\mathcal{K})$$

$$C_0(\mathcal{K}) \xleftarrow{\partial_0} C_1(\mathcal{K}) \xleftarrow{\partial_1} C_2(\mathcal{K}) \xleftarrow{\partial_2} \cdots$$

  - $C_\bullet(\mathcal{K})$ is a chain complex: $\partial^2 = 0$.

- Coboundary operator on cochains:

$$\delta : C^p(\mathcal{K}) \to C^{p+1}(\mathcal{K})$$

$$C^0(\mathcal{K}) \xrightarrow{\delta^0} C^1(\mathcal{K}) \xrightarrow{\delta^1} C^2(\mathcal{K}) \xrightarrow{\delta^2} \cdots$$

  - $C^\bullet(\mathcal{K})$ is a cochain complex: $\delta^2 = 0$.

Discrete Stokes theorem

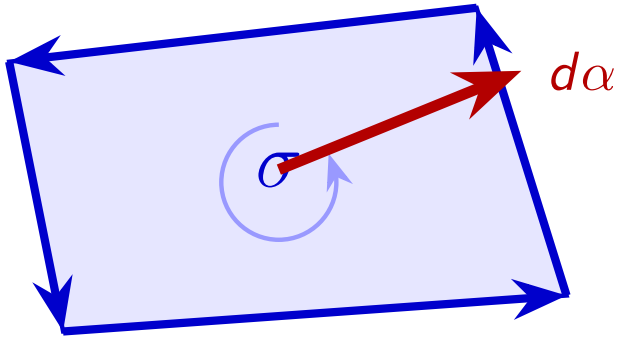
- Combinatorial analogue of **Stokes' theorem**:

$$\langle \delta\varphi, c \rangle = \langle \varphi, \partial c \rangle$$

- Boundary/coboundary are dual: "integration by parts" is algebraic.
- Conservation and compatibility become identities in the cochain complex.
- This is the structural reason cochains are the right level of abstraction for transport.

CIRCULATION

$$\oint_{\partial\sigma} \alpha$$



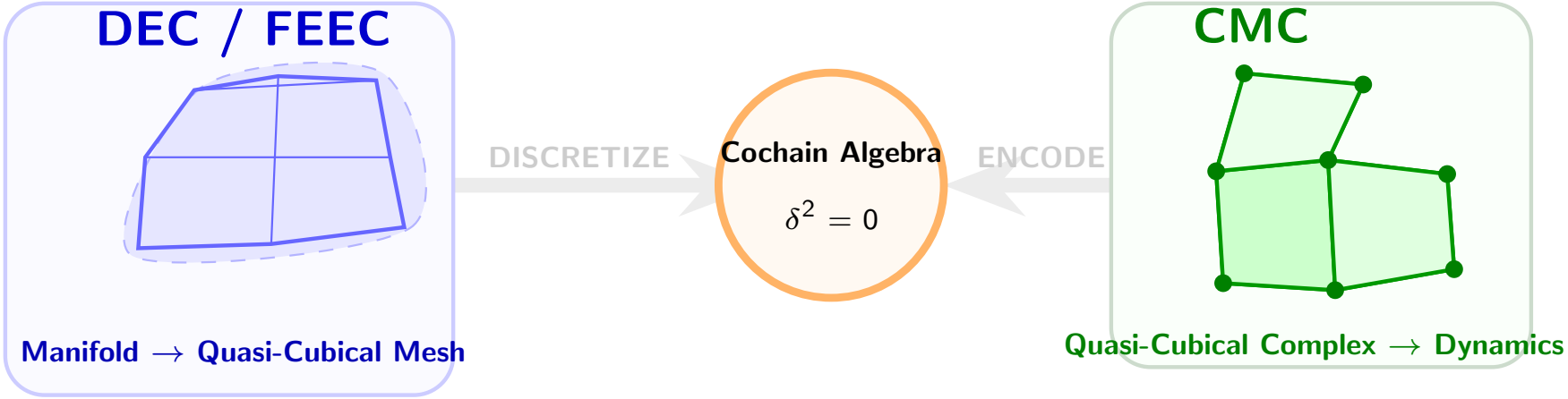
$$=$$

TOTAL FLUX

$$\int_{\sigma} d\alpha$$

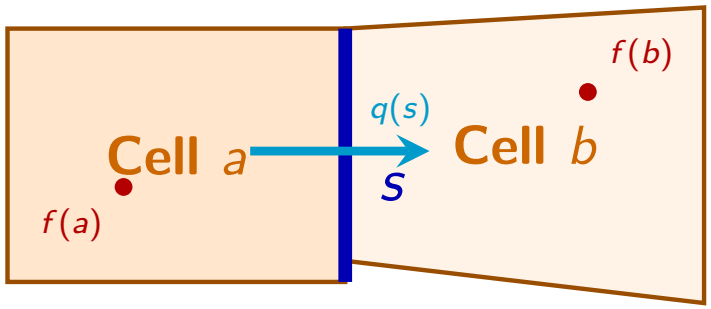
Relation to discrete exterior calculus and FEEC

- **Shared:** cochains, coboundary.
- **Different emphasis:** not a discretisation of a given manifold; $\mathcal{K}$ is primary and may evolve.
- **Different challenge:** vector-valued quantities with *non-uniform fibres* and transport as extra structure.



Scalar quantities on the complex

- Extensive quantities attach naturally to top-dimensional cells (or chosen p-cells).
- Fluxes attach to codimension-one cells (interfaces), consistent with incidence.
- Balance laws are expressed using coboundary: **outflow is a coboundary of flux**.



Combinatorial Incidence

$$[a : s] = +1 \quad [b : s] = -1$$

Intrinsic Balance

$$\sum_{s \in \partial a} [a : s]\, q(s) = f(a)$$

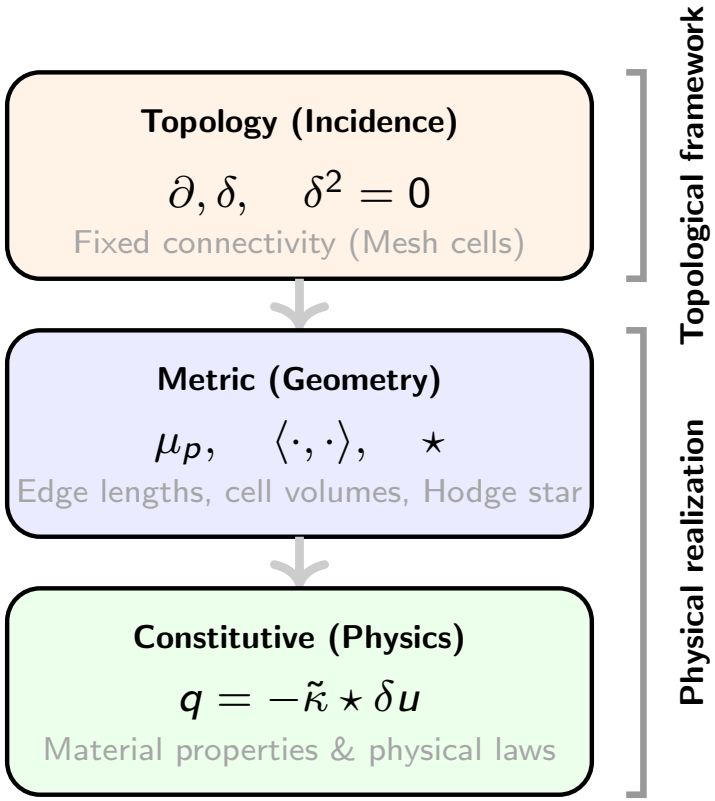
Metric and Constitutive Structure

Metric is introduced *after* topology

- **Topology:** incidence $\Rightarrow \partial, \delta$
- **Metric:** assign measures to cells, then define inner products
- **Geometry:** constitutive maps use the metric layer

Inner products on cochain spaces

- **Cell measures (weights):** choose $\mu_p : \mathcal{K}_p \to \mathbb{R}^+$.
- **Inner product** on $C^p(\mathcal{K})$ defined for given μ_p.
- **Adjoint coboundary** $\delta^\star : C^{p+1}(\mathcal{K}) \to C^p(\mathcal{K}$ defined by $\langle \delta\sigma, \tau \rangle_{p+1} = \langle \sigma, \delta^\star \tau \rangle_p$.

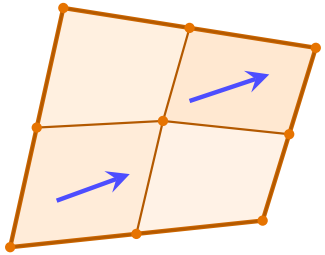


Applications of the scalar framework ▸ Berbatov 2025

Strong, primal weak and mixed weak formulations developed, and used for

1. **Diffusion on heterogeneous complexes**
   mass diffusion / heat conduction / charge transport on irregular cells
2. **Flow in porous and fractured media**
   flux through $(D-1)$-cells; interfaces model cracks and pores ▸ Liu 2025 ▸ Liu 2026
3. **Coupled thermo–chemical transport**
   multiple scalar species + reaction/production terms on D-cells
4. **Variational initial–boundary problems**
   primal & mixed forms; efficient solves via diagonal stiffness in the mixed form

Quasi-Cubical Complex K



Intrinsic Fluxes $\psi \in \Omega^1(K)$

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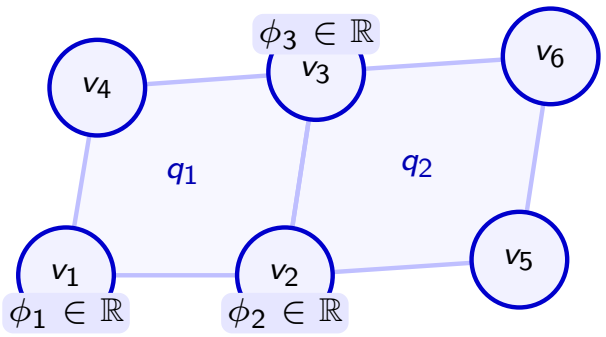
- Mass and heat diffusion
- Fractured media flow
- Thermo-chemical flow
- Variational IBVP

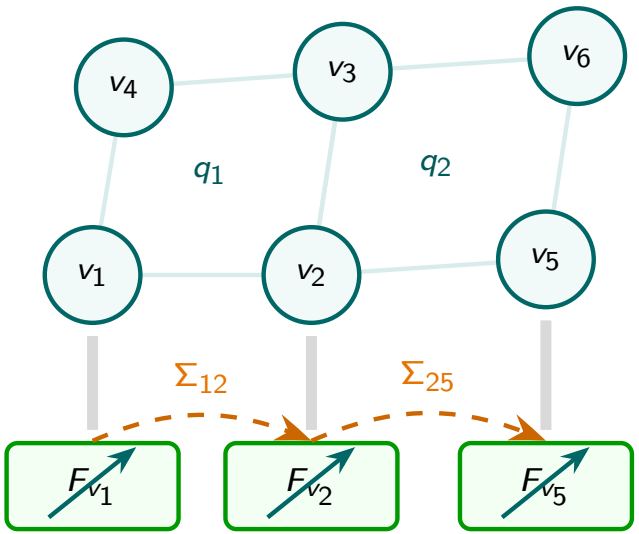
Scalars are not sufficient

- Mechanics requires vector/covector-type quantities (forces, momenta, displacements).
- In a combinatorial setting there is no ambient tangent bundle by default.
- Therefore vectors must live in **fibres** attached to cells/vertices; transport is additional structure.

Scalar field on a Quasi-cubical complex



Vector-bundle on a Quasi-cubical complex

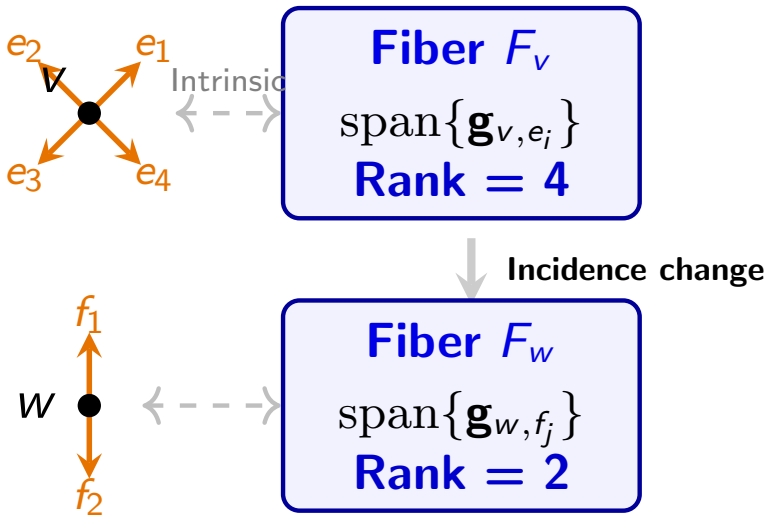


DISCRETE GEOMETRY

- Base cells are quadrilateral (quasi-cubical) cells q_i
- Values or vectors live on vertices / attached fibers F_v
- Transport Σ acts intrinsically along neighboring cells

A bundle over the 0-skeleton (with varying fibre rank)

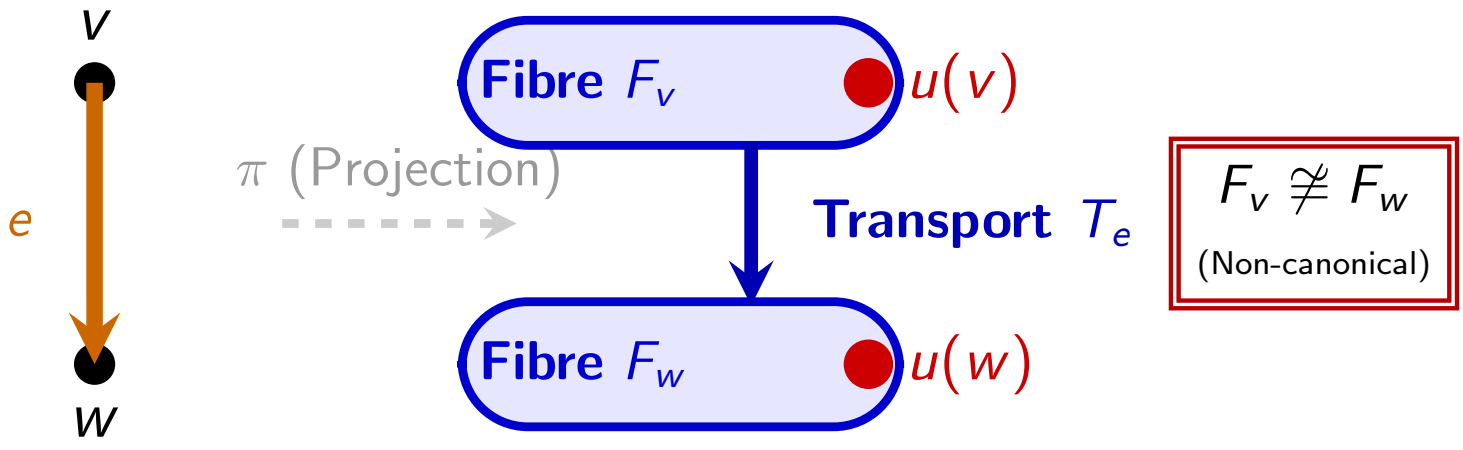
- Attach a vector space F_v to each vertex v (a "bundle" over $\mathcal{K}_0$).
- F_v is generated by edges incident to v (star of v).
- **Key difficulty:** $\dim F_v$ varies with valence; there is no canonical identification $F_v \cong F_w$.



Vector space dimensions are determined by local topology.

Vector-valued 0-forms as sections

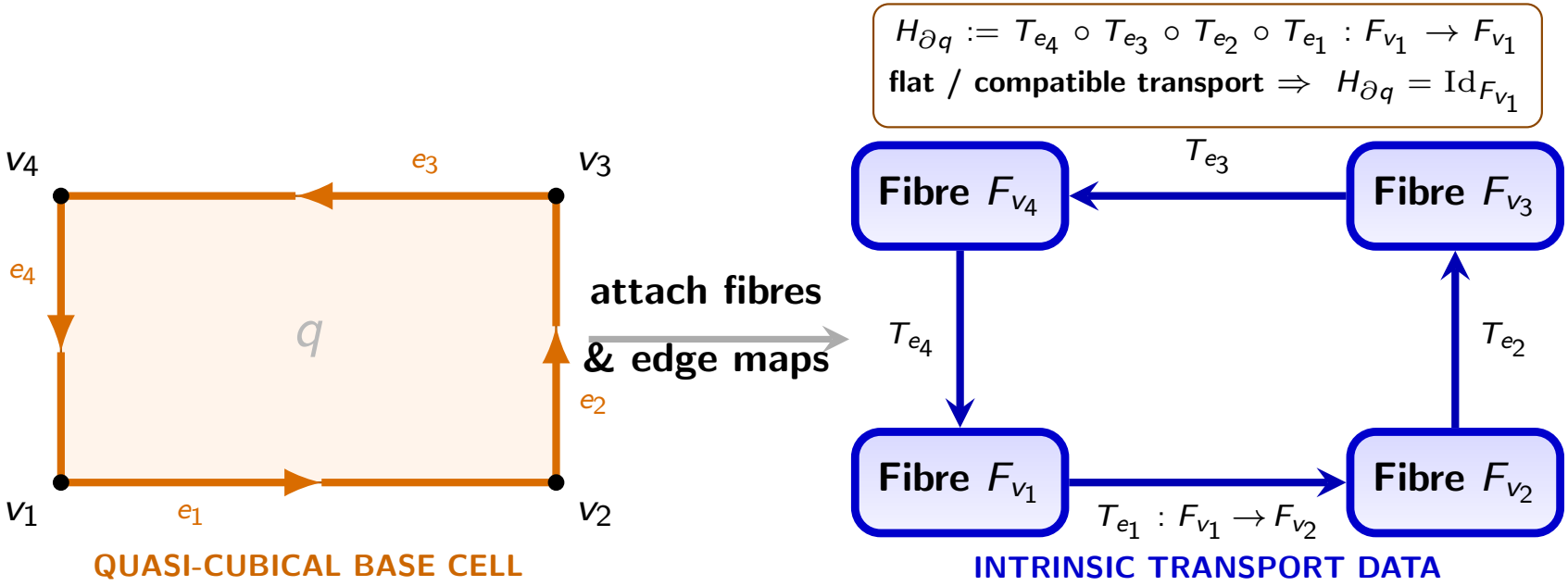
- A vector field is a choice of element $u(v) \in F_v$ at each vertex.
- Comparing $u(v)$ and $u(w)$ requires a transport rule (not automatic).
- This is the point where "connection-like" structure becomes necessary.



Base Complex

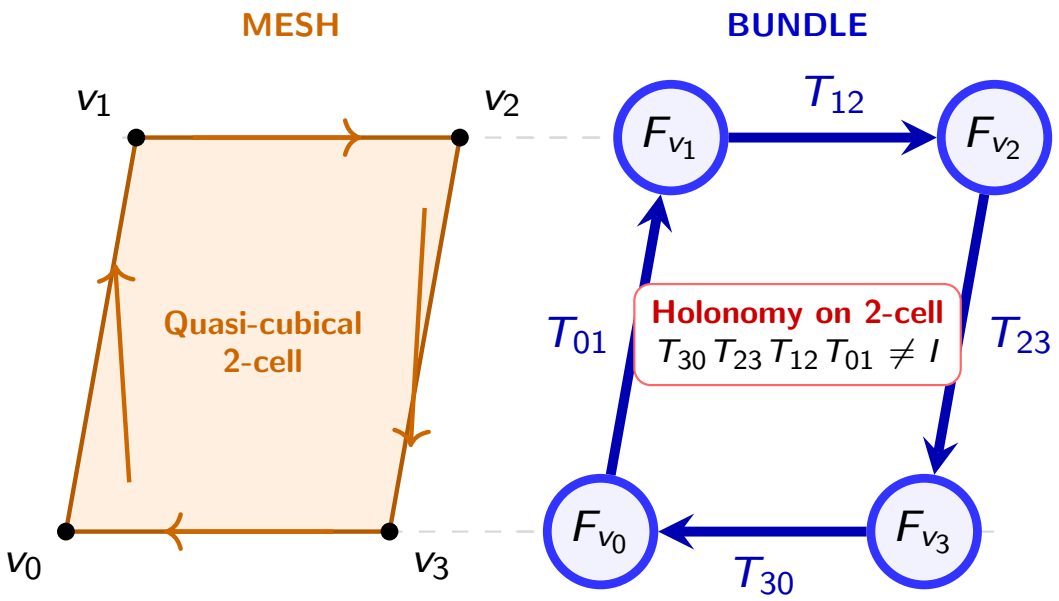
Transport along edges: a groupoid viewpoint

- The 1-skeleton paths form a **path groupoid** (objects = vertices, morphisms = paths).
- A connection is an assignment of linear maps to generating morphisms (edges), extended by composition.
- Curvature appears when the 2-cells impose relations that are not respected by the transport (holonomy defect).



Curvature as holonomy defect (primary notion)

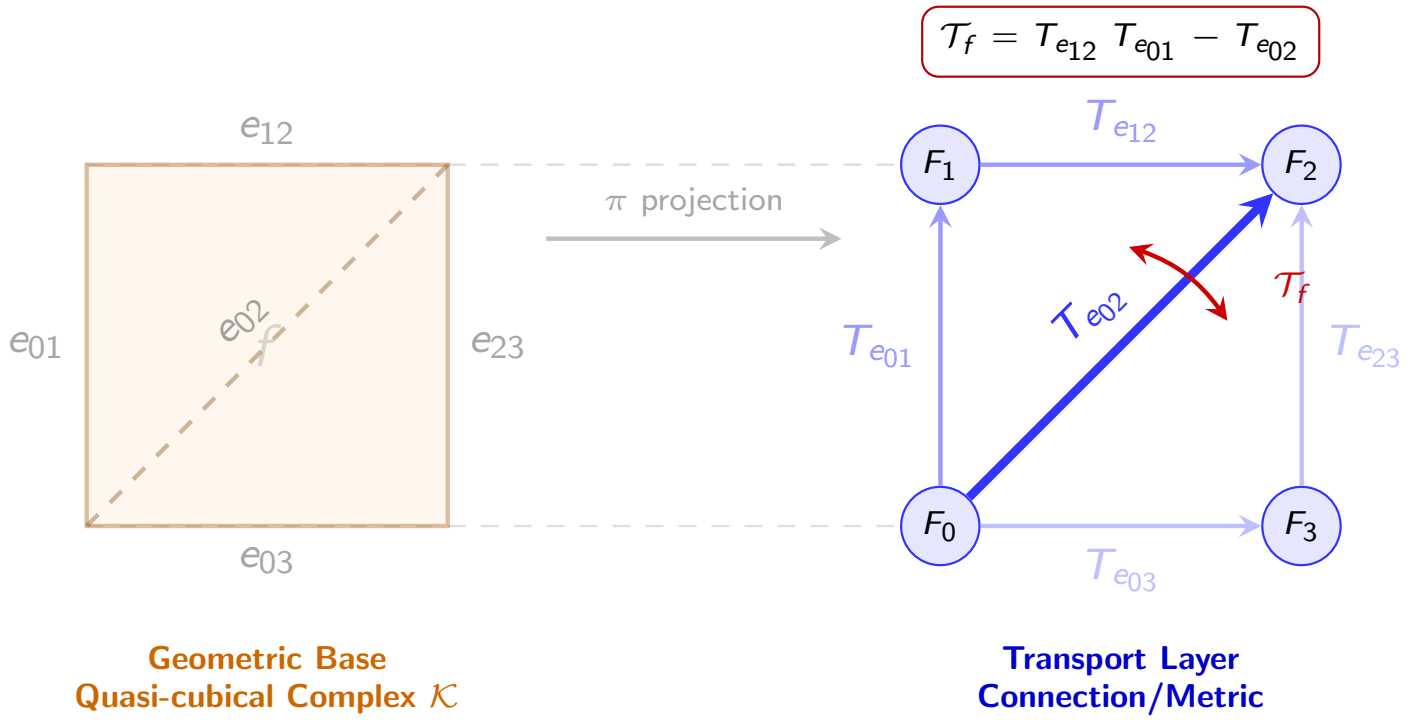
- A 2-cell determines a closed boundary walk in the 1-skeleton.
- Parallel transport around that loop yields an endomorphism of a fibre (holonomy).
- **Curvature:** deviation of holonomy from identity (or from a reference transport).



Transport around a quasi-cubical face detects intrinsic curvature

A torsion-like obstruction (compatibility defect)

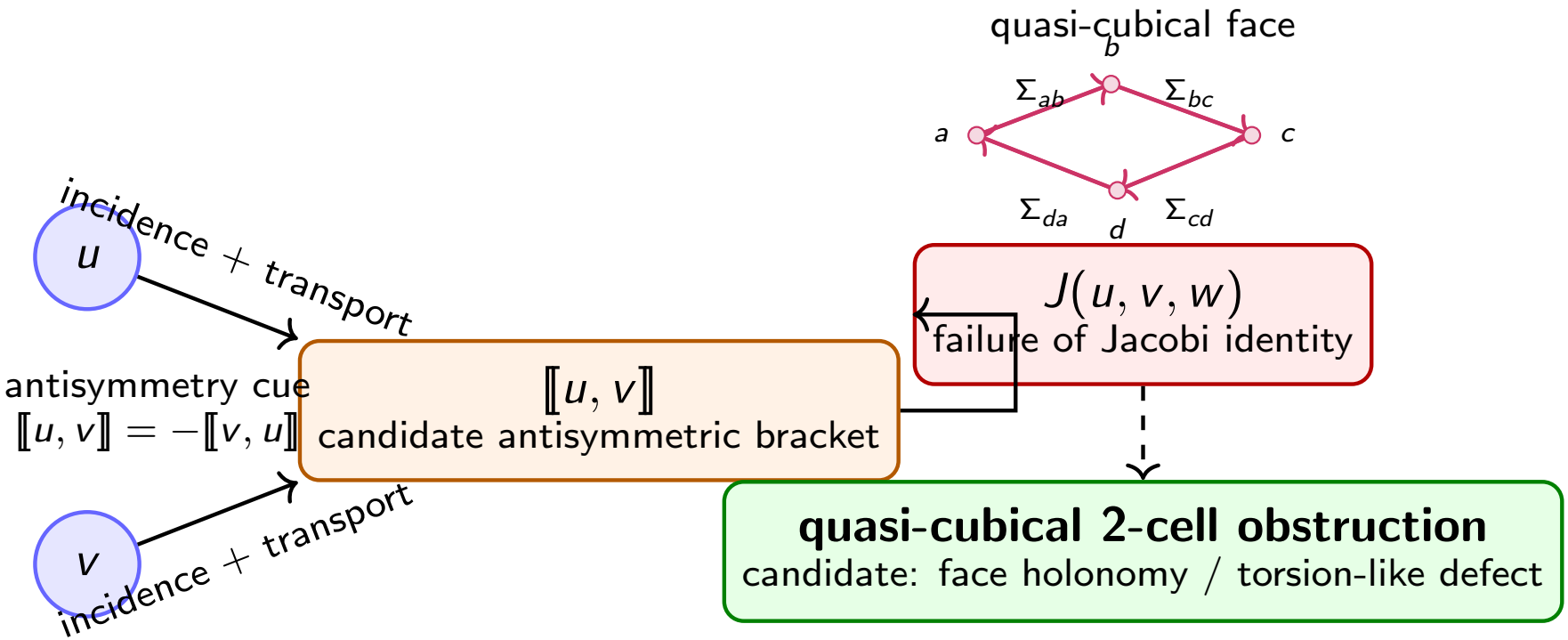
- Measures failure of a chosen "direct" transport to agree with composed transport along a 2-step path.
- Interpretable as **compatibility defect** tied to incidence + transport choices.
- If you want to call it torsion: specify the additional structure (frame/soldering) it refers to.



An antisymmetric bracket on sections (current work)

Berbatov 2026

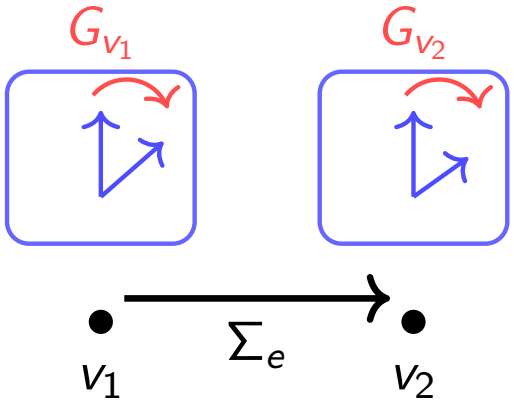
- Define an antisymmetric bilinear operation on vector-valued 0-forms using incidence + transport.
- The Jacobiator provides an **algebraic obstruction** (generally nonzero).
- This obstruction is a candidate invariant; its precise relation to curvature/torsion is an open question.



Invariants and gauge (what should be classified?)

- Gauge change: re-choose bases in each F_v (vertex-wise change of frames).
- Holonomy classes around 2-cells: conjugacy-type invariants under gauge.
- Cohomological data (scalar sector) vs holonomy/obstruction data (vector sector).

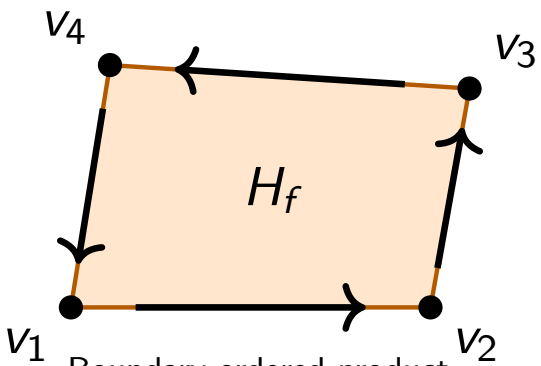
Gauge change



Re-choose basis in each fibre

$\Sigma_e \mapsto G_{v_2} \Sigma_e G_{v_1}^{-1}$

Holonomy around a square 2-cell



Boundary-ordered product

$H_f = \Sigma_{e_4} \Sigma_{e_3} \Sigma_{e_2} \Sigma_{e_1}$

What is classified?

Scalar / abelian sector
cohomological data

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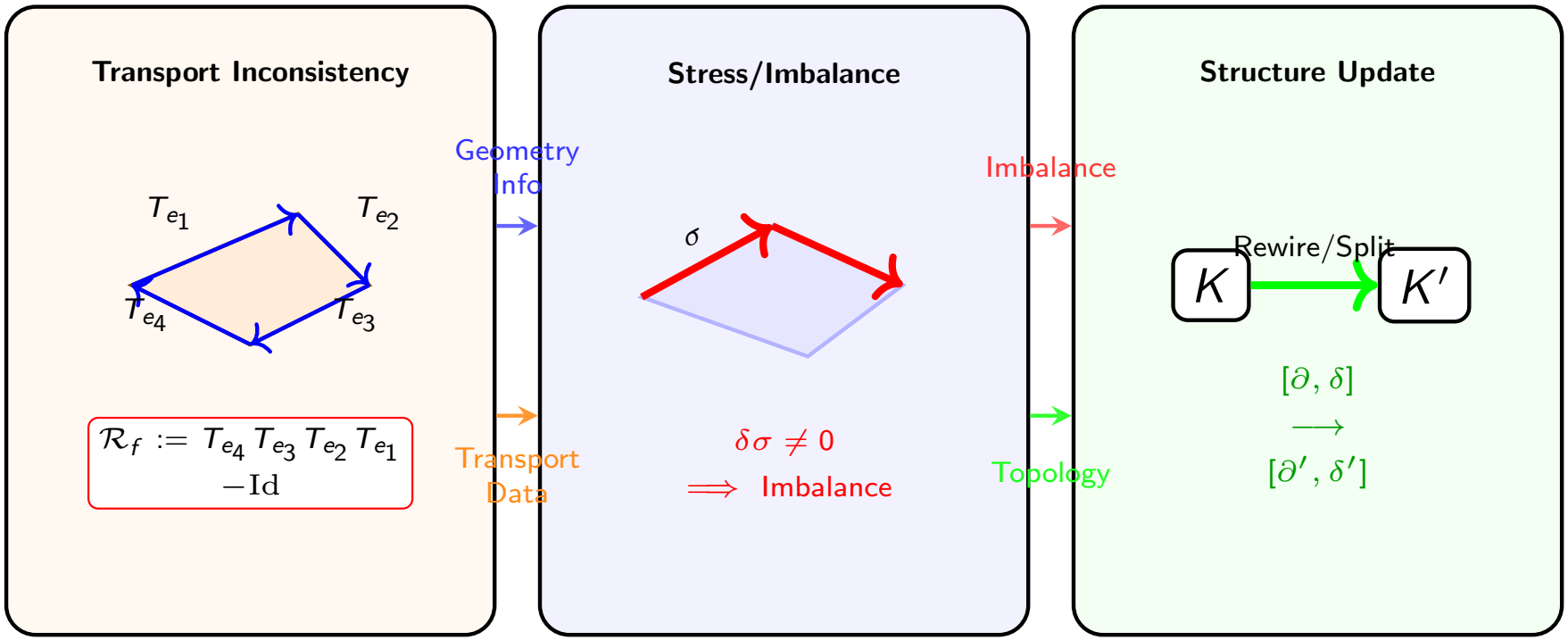
Transport sector
holonomy up to conjugacy

For scalar/abelian transport one expects cohomological invariants, whereas for general bundle-valued transport the intrinsic gauge-invariant

Why mechanics cares (brief motivation)

- Forces/defects concentrate on lower-dimensional features (interfaces, junctions): naturally "cell-supported".
- Transport inconsistencies (holonomy/compatibility defects) suggest a geometric language for internal structure.
- This is motivation only; the mathematical development is the focus of this seminar.

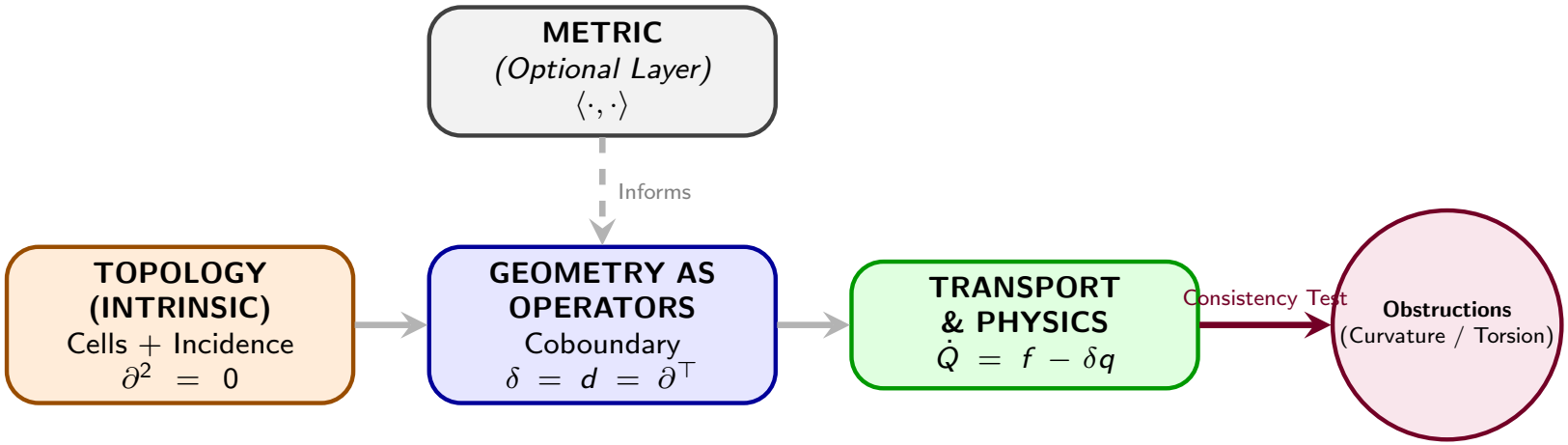


Open mathematical questions

- What is the right class of admissible edge transports when fibre ranks vary?
- What are the right curvature/holonomy invariants under gauge and refinement?
- How should torsion be defined (what additional structure is required)?
- How do these structures behave under the evolution of K (rewiring, splitting, refinement)?

Closing

- Cochains provide an intrinsic calculus on a finite complex (topology first, metric optional).
- Vector-valued quantities force us to confront transport/connection as *extra structure*.
- **Closing question:** which obstructions (holonomy, Jacobiator, torsion-like defects) are the right "geometry" of a combinatorial system?



Incidence first → Metric later
→ Transport drives meaning

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